



MySip.ch

Users Manual

**How to build a test environment
Version 1.0**

**Issued by
DS MS, Software house
Albisriederstr. 245, CH-8047 Zurich**

Copyright © Siemens Schweiz AG 2004

All Rights Reserved.

SIEMENS SCHWEIZ AKTIENGESELLSCHAFT

**Author: P.Mächler
Translator:**

DS MS 1

In addition to the authors named on the cover page, the following persons have collaborated on this document:

Shaun Baker
Markus von Arx

DS MS
DS MS

The document comprises 20 pages, all pages have issue no 01.

The document is based on template RSPEC.DOT.

This issue was last saved on 07.01.2004 09:03.

This document was edited with MS WinWord version 8.0b.

The filename of the main document file was

J:\PROJECTS\ID\DS_MS\SIP\General_Documents\TestInstallation_of_SCSCClient.doc.

0 General Information	4
0.1 Issue Control	4
0.2 History	4
0.3 References	4
0.4 Glossary and Abbreviations.....	4
0.5 Keyword/Descriptor	5
0.6 List of Figures and Tables	5
1 Introduction	6
1.1 Scope.....	6
2 Build a standalone test environment in your LAN	7
2.1 Basic Architecture.....	7
2.2 Installation	7
2.3 Configuration.....	8
2.4 Configure Proxy	10
2.4.1 Domain name.....	10
2.5 Configure the SCS-Client.....	11
3 Build a test system in the Internet	13
3.1 Basic Architecture.....	13
3.1.1 Limitations on the actual SCS-Client	13
3.2 Configure the SCS-Client.....	14
4 Build a test system over the Internet with VPN	15
4.1 Basic Architecture.....	15
4.1.1 Limitations on the actual SCS-Client	15
4.2 Configure the SCS-Client.....	16
5 Adding a gateway to the environment	17
5.1 Configuration.....	17
5.1.1 Configuration of the gateway	18
5.1.2 Configuration of the gatekeeper.....	20

0 General Information

0.1 Issue Control

The document comprises 20 pages, all pages have issue no XX.

0.2 History

Issue	Date	Reason for Changes
02	01.12.2003	Internet test environment added
01	20.11.2003	Creation

Table 1: History

0.3 References

/1/ Hicom Xpress IWU
 Requirement Specification
 P00914-A1000-V020-01-0026

0.4 Glossary and Abbreviations

DTMF	D ual T one M ultiple F requency (Touch tone)
ISDN	I ntegrated S ervices D igital N etwork
IWU	I nter W orking U nit
PBX	P rivate B ranch eX change
PSTN	P ublic S witched T elephone N etwork

0.5 Keyword/Descriptor

0.6 List of Figures and Tables

Table 1: History	4
------------------------	---

1 Introduction

1.1 Scope

This document contains a short description on how to build a test environment for the SIP-Products of Siemens Switzerland Ltd.

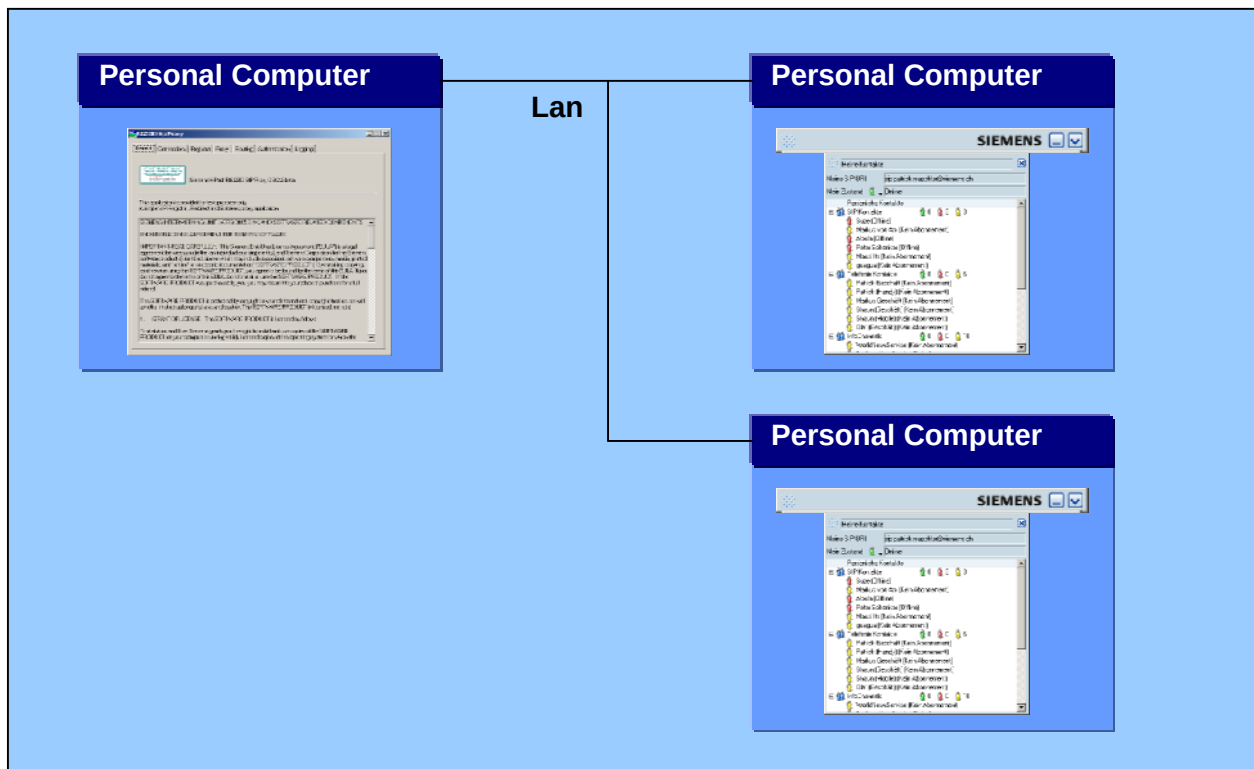
2 Build a standalone test environment in your LAN

This chapter describes how to install the necessary SIP components for testing in a standalone LAN environment.

2.1 Basic Architecture

The smallest configuration you can build needs 3 machines:

- **1 Machine for the Proxy/Registrar**
You can download this software under <http://www.mysip.ch>, called **SIPProxy**
- **2 Machines for two clients to communicate**
The software can be found under <http://www.mysip.ch>, called **SCS-Client**



Please be aware that you additionally need an ethernet switch (or hub) for this configuration.

2.2 Installation

There is documentation on <http://www.mysip.ch> that explains in detail how to install the above mentioned software.

2.3 Configuration

The easiest way to describe the configuration is by example. Therefore we directly define IP addresses for a test scenario. Anyone familiar with DNS names could configure and define for DNS. But to keep the example as simple as possible, private space IP addresses are used here.

IP address	Machine
10.12.1.4	Proxy /Registrar
10.12.1.135	Client 1
10.12.1.136	Client 2

To understand what we have to configure, a short digression on how SIP works in general is necessary.

Think about telephony:

- *You just need to know the number of the user you want to call. The system itself knows on which port this number is connected*

Different with SIP:

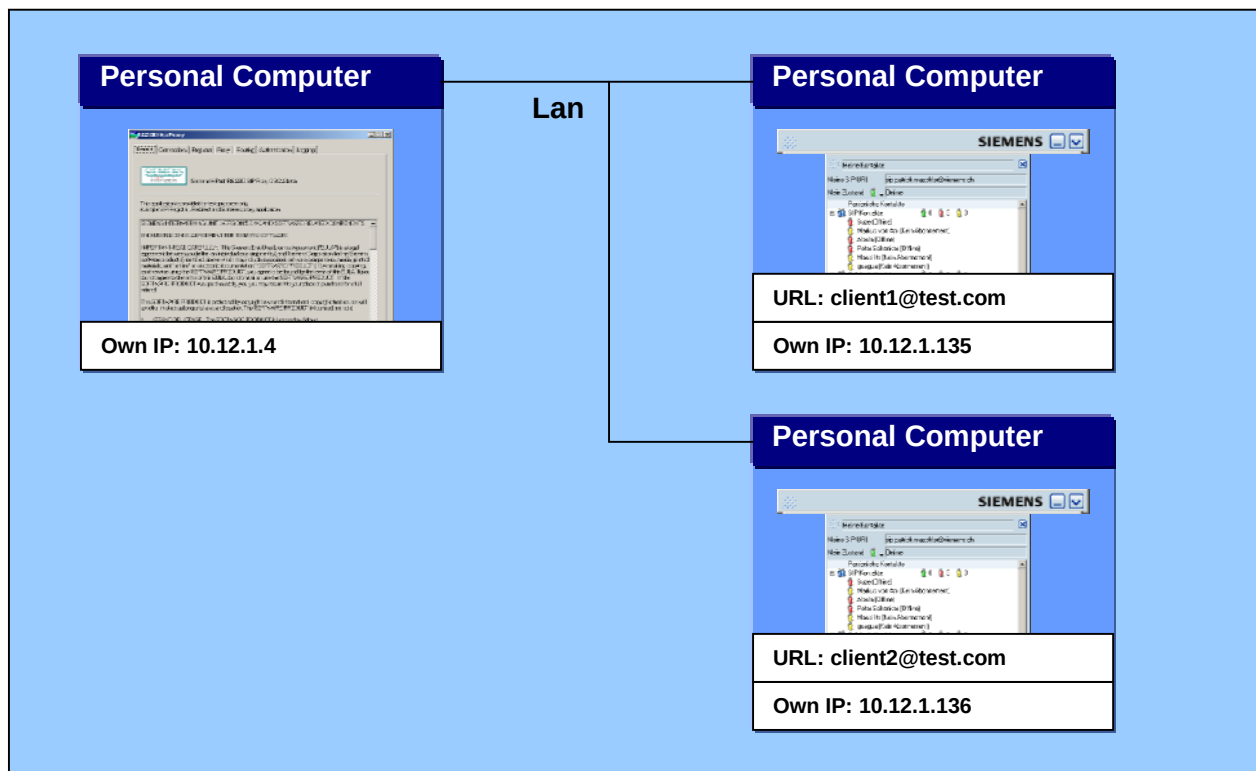
- *Wherever you log in on a machine and start your client, you want to be contactable by other users.*
- *Therefore someone has to know if and where you are contactable at the current moment*
- *In a SIP environment, the proxy/registrar is responsible for that*
- *Therefore the only address your client needs to know is the proxy/registrar.*
- *The registrar/proxy on the other hand has to know who you are and which domains it has to support*

That means, that each user needs a SIP URL (which looks like an e-mail address and in most cases is the same)

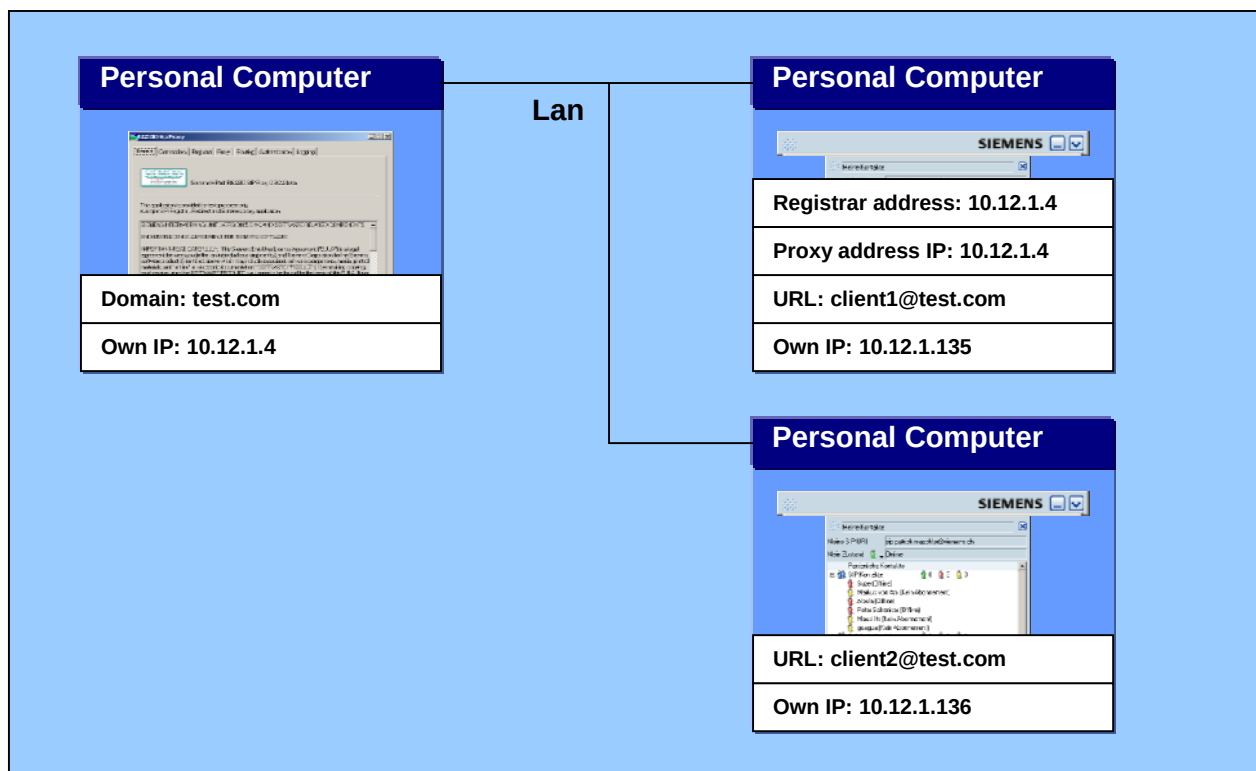
So let's choose two Client user names, but be aware that this name can change, depending on the user that logs into the computer with his account.

PC User	URL
Client1	Client1@test.com
Client2	Client1@test.com

Our configuration now looks like this:



We also know the proxy / registrar address, which is **10.12.1.4**. This allows the user of the client to tell the others that he's online. Also we added for our test environment the domain **test.com**.



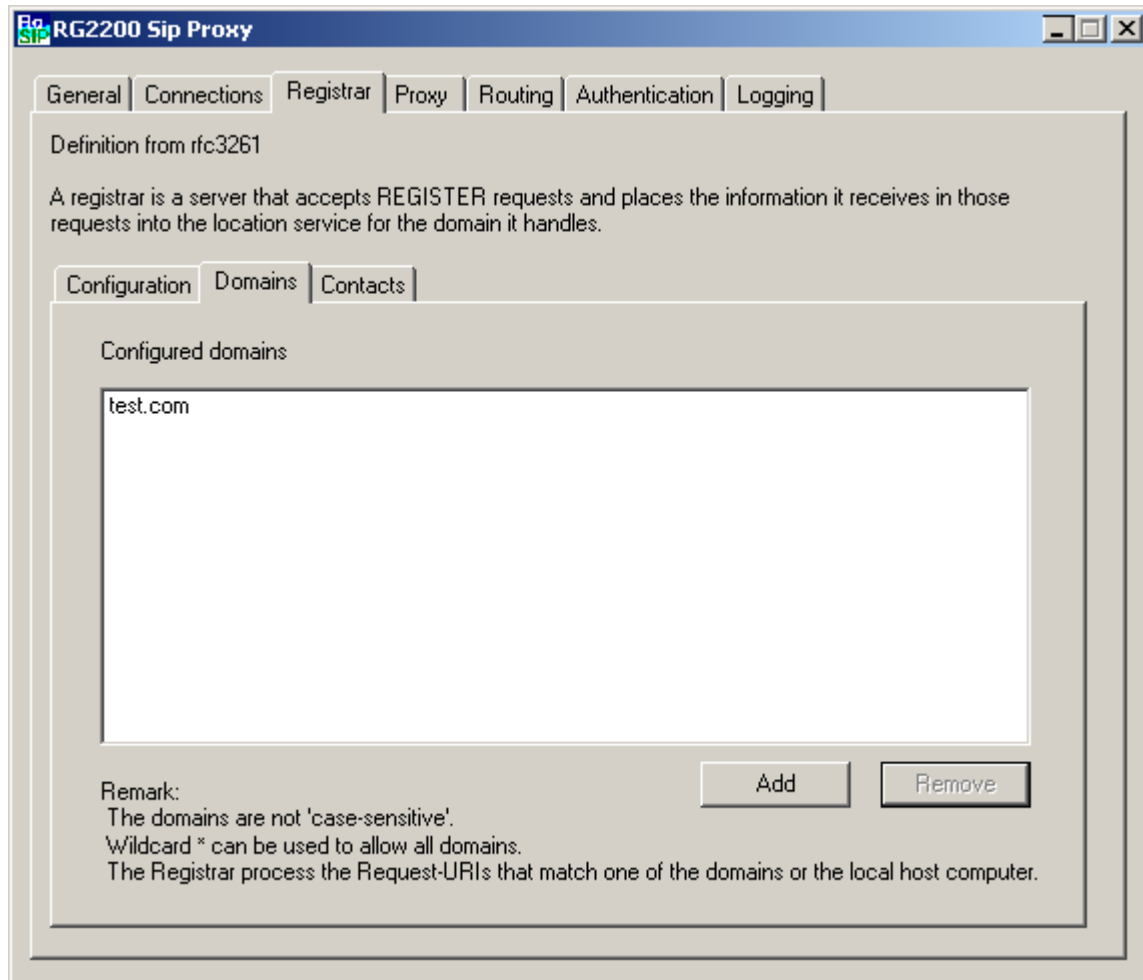
And that's all the information we need to configure for the software.

2.4 Configure Proxy

As we now know which parameters to set, we now need to know where to set them.

After starting up the proxy, configure the following parameters:

2.4.1 Domain name



Add the domain name you chose (here 'test.com'). The proxy discovers its own IP Address, so no further configuration is necessary (for the moment).

2.5 Configure the SCS-Client

During first startup, you're able to fill in all parameters needed. I guess you just tried something that didn't work; and now you downloaded this document to get it running (I do that normally too ☺). Therefore I will explain how to configure it using the configuration dialog:

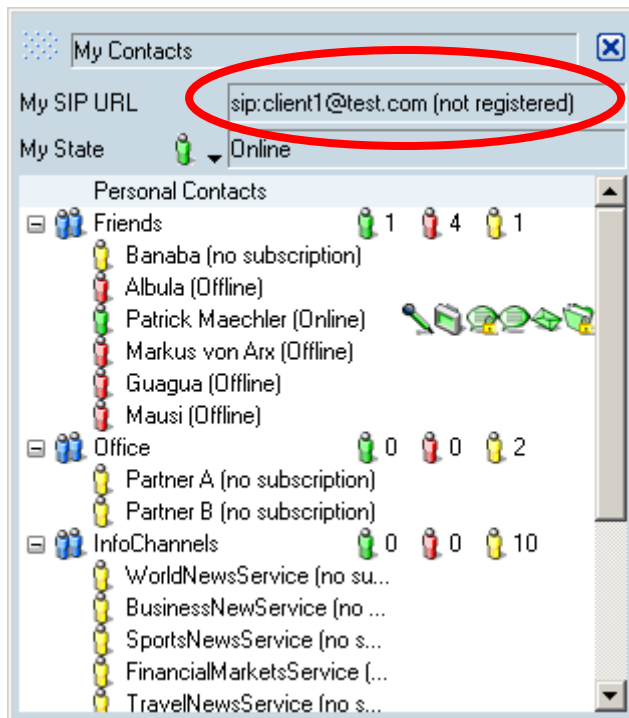
Open in the main menu "Configuration" (If you are not familiar with the client, press "F1" for the online help of the SCS-Client) and enter the following values:

The screenshot shows the 'Configuration' dialog box with the following settings:

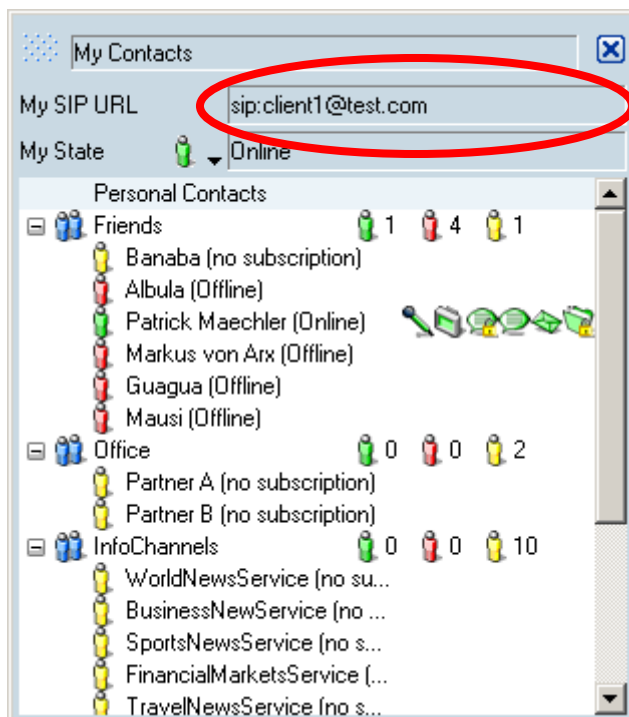
- Presence** tab is selected.
- GUI** sub-tab is selected.
- User** section:
 - Display Name: Client1
 - SIP URL: sip:client1@test.com
 - Email Address: client1@test.com
 - Physical Address: sip:10.12.1.135:5061
- Server** section:
 - ☒ Enable SIP Registrar: sip:10.12.1.4
 - ☒ Enable SIP Proxy: sip:10.12.1.4
- Realm Passwords** section:
 - Table with 2 columns: realm, username.
 - Remove button.
- Buttons:** OK, Cancel.

That's it for the client.

If you configured it wrong, the following message will occur on the SCS-Client Contact box:



If you did it correct:



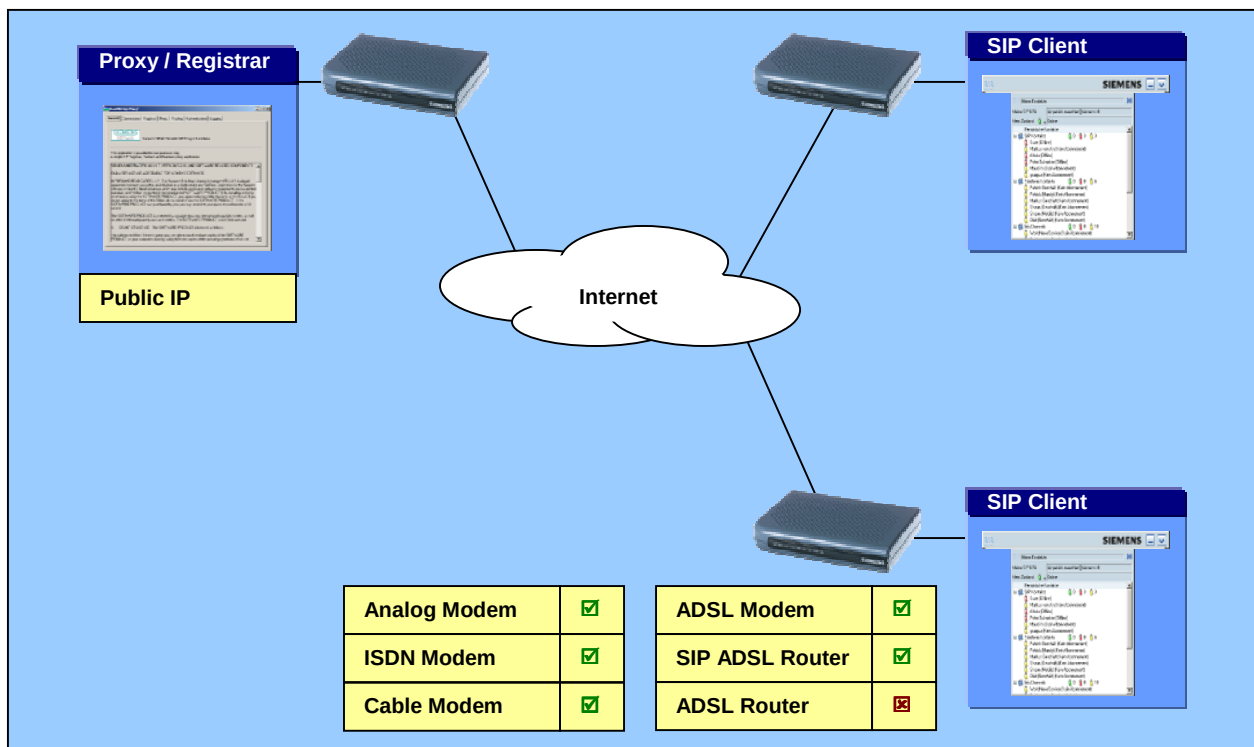
3 Build a test system in the Internet

This chapter describes how to install the necessary SIP components for testing in the Internet.

3.1 Basic Architecture

The smallest configuration you can build needs 3 machines:

- **1 Machine for the Proxy/Registrar**
- **2 Machines for two clients to communicate**



If you don't have a public IP-address for the proxy, you can test with the proxy **sipproxy.mysip.ch**. Please be aware that this proxy is for testing purpose only, limited to 100 users and no guaranteed uptime.

3.1.1 Limitations on the actual SCS-Client

The actual SCS-Client doesn't support STUN or uPNP that is necessary to handle NAT on the ADSL Routers. There are, however, a few suppliers - such as INTERTEX - offering SIP capable ADSL Routers.

In the near future, the SCS-Client will support uPNP, supported by the main suppliers of ADSL Routers (Zyxel, Siemens, etc.).

3.2 Configure the SCS-Client

Open in the main menu “Configuration” (If you are not familiar with the client, press “F1” for the online help of the SCS-Client) and enter the following values:

The Configuration dialog box is shown with the following settings:

- User:**
 - Display Name: Patrick Mächler
 - SIP URL: sip:patrick.maechler@mysip.ch
 - Email Address: patrick.maechler@mysip.ch
 - Physical Address: sip:139.16.184.233:5061
- Server:**
 - ☒ Enable SIP Registrar: sip:sipproxy.mysip.ch
 - ☒ Enable SIP Proxy: sip:sipproxy.mysip.ch
- Realm Passwords:**

realm	username
-------	----------

Remove

Please enter your own

- Display Name
- SIP URL
- Email Address

Check configuration as described in chapter 2.5.

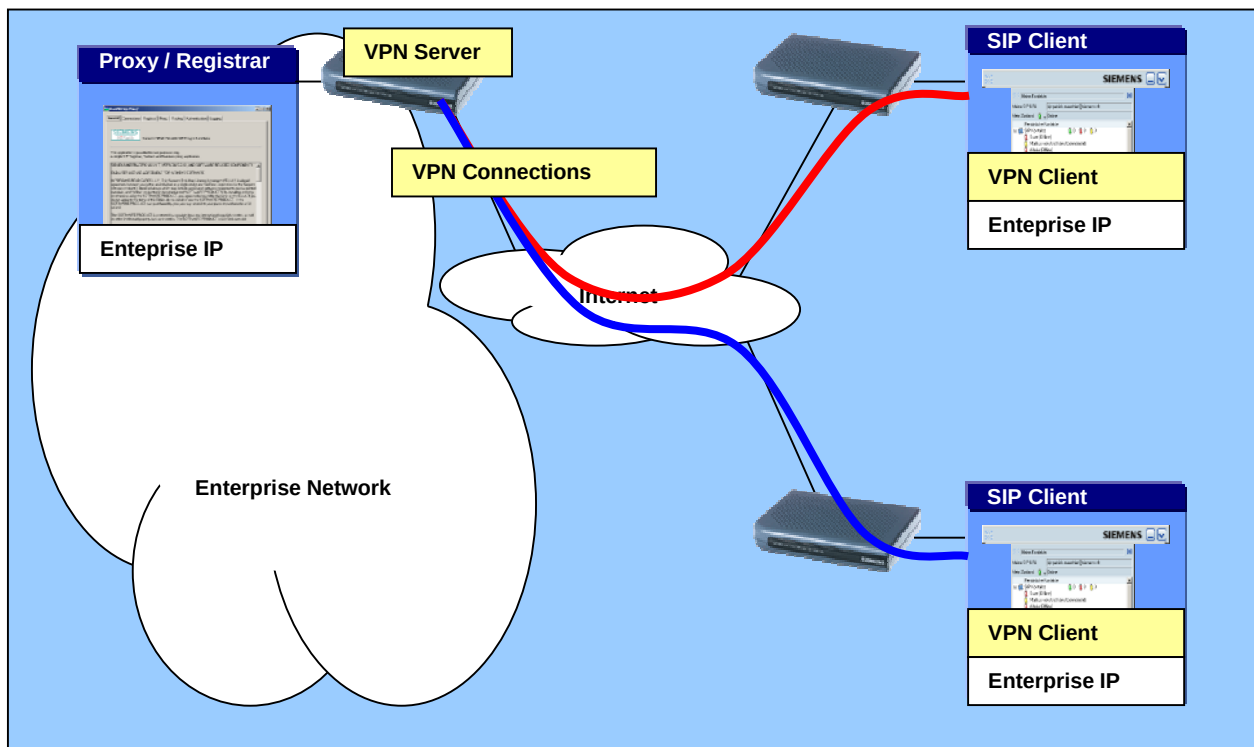
4 Build a test system over the Internet with VPN

This chapter describes how to install the necessary SIP components for testing over the Internet with VPN (Virtual Private Network). This architecture is mainly used in the enterprise environment, where home workers want to access the corporate network. VPN has the main advantage, that is independent of the transport network. Therefore we don't have to care about NAT problems. Disadvantage is, that all connections go over the Enterprise network, which means large bandwidth usage on that point (no direct connection between SIP Client 1 & 2 over the internet possible)

4.1 Basic Architecture

The smallest configuration you can build needs 3 machines:

- 1 Machine for the Proxy/Registrar
- 2 Machines for two clients to communicate



4.2 Configure the SCS-Client

Open in the main menu “Configuration” (If you are not familiar with the client, press “F1” for the online help of the SCS-Client) and enter the following values:

Configuration

Presence | Presence Filters | Audio/Video

GUI | Colors | Info-Channels | Boilerplates | Account

User

Display Name: Patrick Mächler

SIP URL: sip:patrick.maechler@mysip.ch

Email Address: patrick.maechler@mysip.ch

Physical Address: sip:139.16.184.233:5061

Server

☒ Enable SIP Registrar: sip:sipproxy.mysip.ch

☒ Enable SIP Proxy: sip:sipproxy.mysip.ch

Realm Passwords

realm	username
-------	----------

Remove

OK Cancel

Please enter your own

- Display Name
- SIP URL
- Email Address

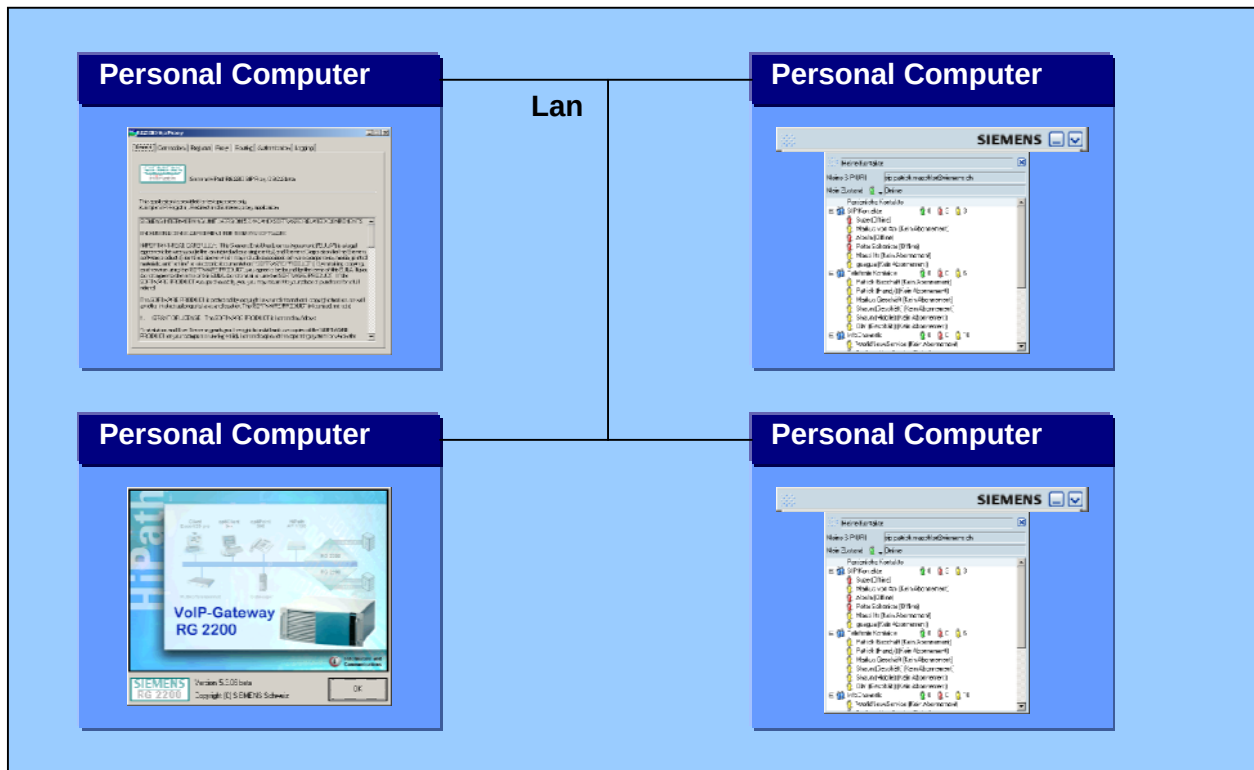
Check configuration as described in chapter 2.5.

Important:

The VPN connection needs to be established before starting the client. Otherwise it won't detect the correct IP-Address for communicating over the VPN connection

5 Adding a gateway to the environment

After successful installation of a proper SIP environment, we now enhance you infrastructure with a gateway to breakout voice to the PSTN



Our local domain name is **test.com**. We now add a new domain called **pstn**. This allows the proxy to discriminate between internal SIP calls and calls to the PSTN

5.1 Configuration

We now add the gateway to the already existing infrastructure.

IP address	Machine
10.12.1.4	Proxy /Registrar
10.12.1.41	Gateway
10.12.1.135	Client 1
10.12.1.136	Client 2

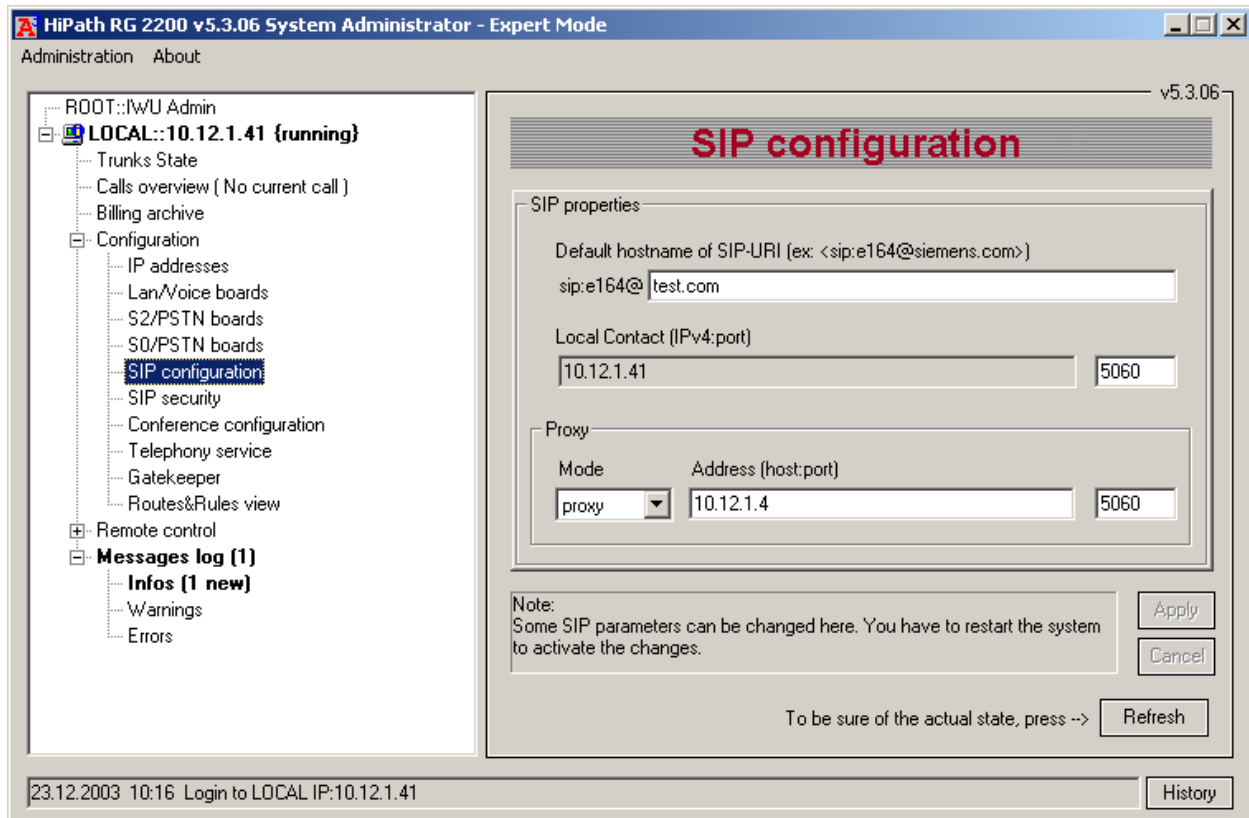
5.1.1 Configuration of the gateway

There exists a detailed User manual for the gateway. For further information, please contact this document. This chapter describes the fundamental configurations.

Please test first the CAPI driver with a tool, delivered with your ISDN card. This helps to prevent searching hours for PSTN access problems.

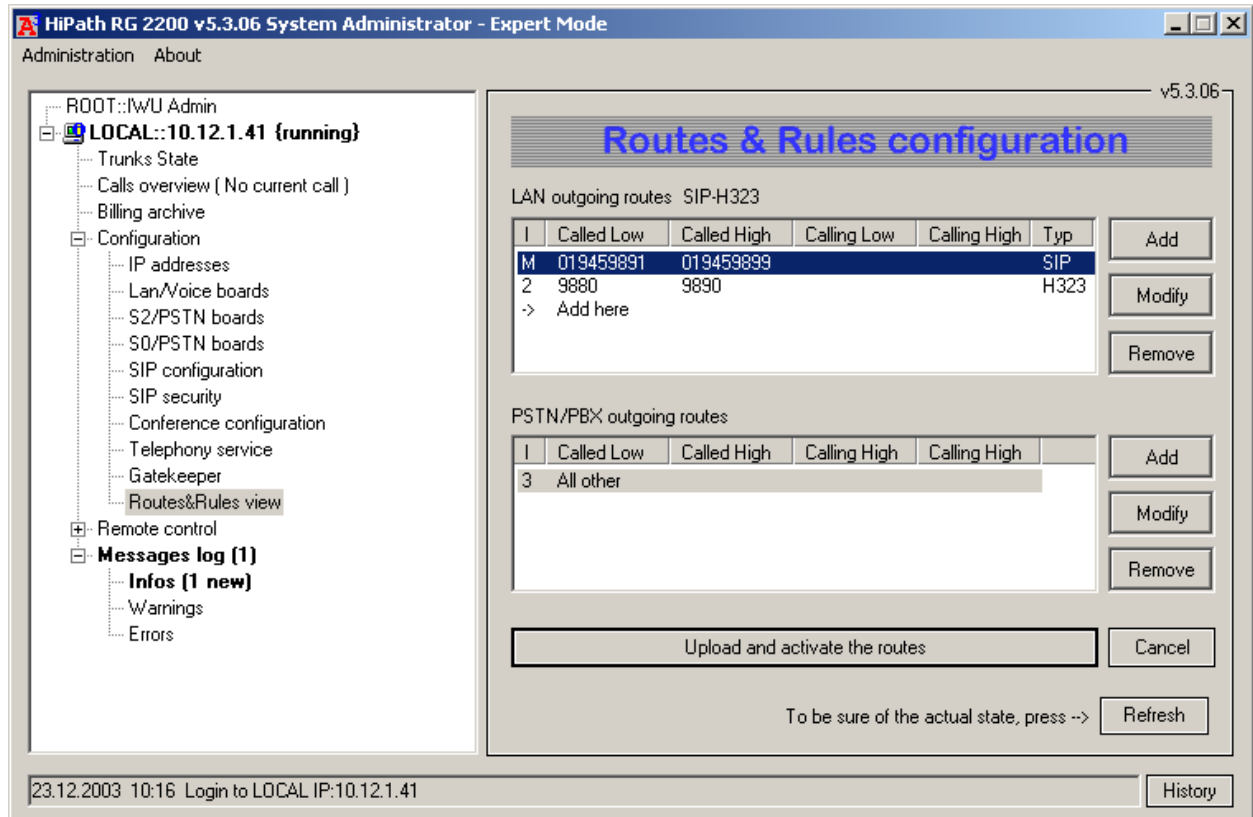
Start the IWUAdmin Tool on the gateway and log in. Default password is **dra**.

First we set the **Configuration** ⇒ **SIP-Configuration**



If you want to receive incoming calls, you have to set route/rules for your MSN numbers (your Telephone Numbers).

Configuration ⇨ Routes / Rules view



In our example, we want to forward the following PSTN Numbers to SIP:

019459891, 019459892, ... , 019459899

It's important to know, how your PSTN provider sends you the number.

If it is in international format, then it looks like this: 004119459891

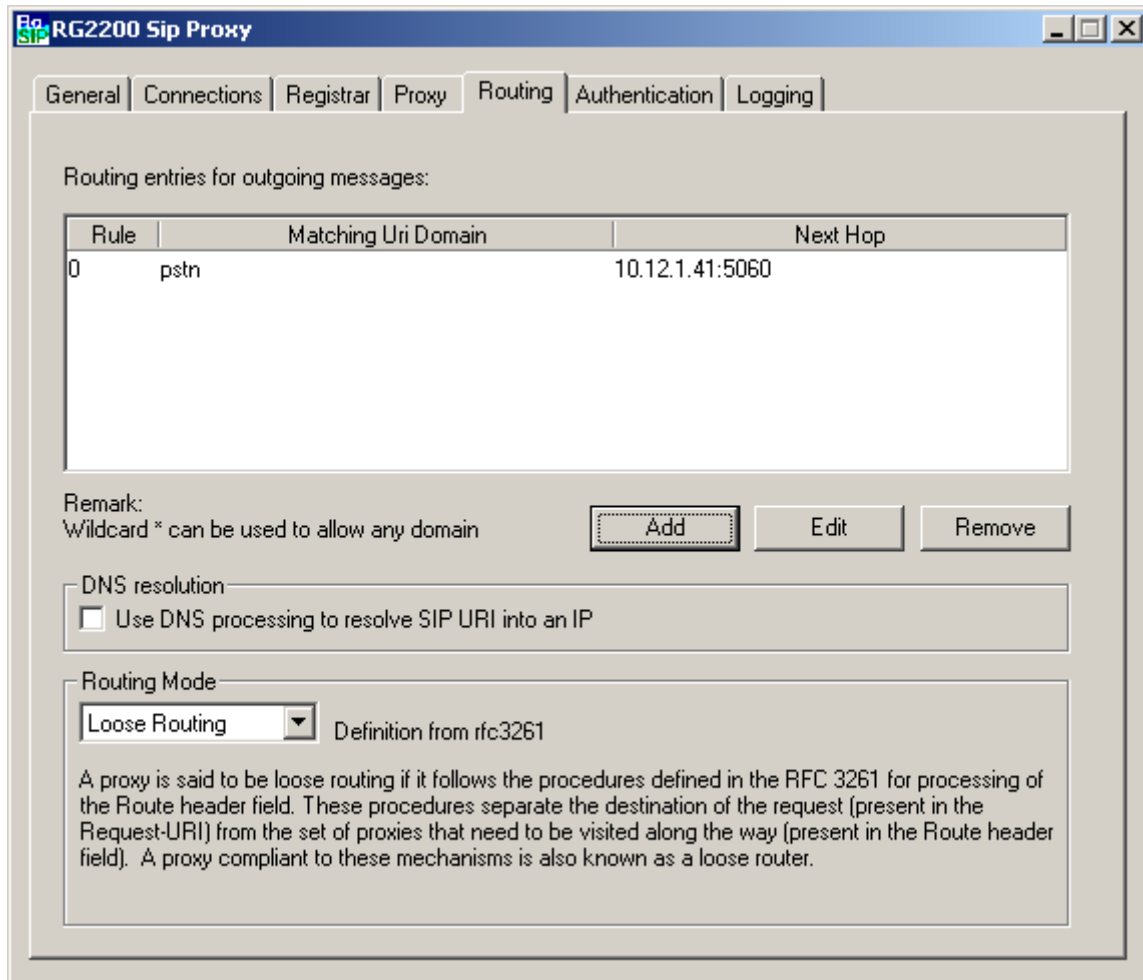
On national format: 019459891

On local format: 9459891

Or just MSN Number: 1

5.1.2 Configuration of the gatekeeper

We now have to add a rule, that the proxy forwards all **<phonenumber>@pstn** will be forwarded to the gateway. The rule looks like this:



If you want to receive calls from the PSTN to a SIP phone, please register a SCS-Client with one of the MSN numbers:

Example: 019459891@test.com